FIITJEE

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST – IV

JEE (Main)-2021

TEST DATE: 16-05-2021

Time Allotted: 3 Hours **Maximum Marks: 300**

General Instructions:

- The test consists of total 90 questions.
- Each subject (PCM) has 30 questions.
- This question paper contains **Three Parts.**
- **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is Mathematics.
- Each part has only two sections: **Section-A and Section-B**.
- **Section – A** : Attempt all questions.
- **Section – B :** Do any five questions out of 10 Questions.

Section-A (01 – 20, 31 – 50, 61 – 80) *contains 60 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

Section-B (21 – 30, 51 – 60, 81 – 90) *contains 30 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+4 marks*** *for correct answer. There is no negative marking.*

Physics

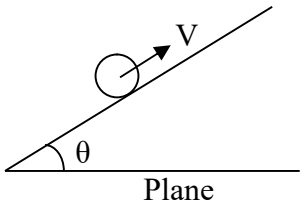
PART – A

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. Two spheres of mass 2M and M are initially at rest at a distance R apart. Due to mutual force of attraction they approach each other. When they are at separation R/2, the acceleration of the centre of mass of spheres would be:
   (A) Zero
   (B) $g\,m/s^2$
   (C) $3g\,m/s^2$
   (D) Data insufficient

2. The sphere is performing rolling up an inclined plane, the frictional force is:
   (A) Directed up the plane
   (B) Directed down the plane
   (C) Zero
   (D) Depending on its velocity



3. A simple pendulum has a time period T_1 when on earth's surface, and T_2 when taken at a height 2R above the earth's surface where R is the radius of earth. The value of T_1/T_2 is:
   (A) $1/9$
   (B) $1/3$
   (C) $\sqrt{3}$
   (D) 9

4. A body is projected up with a velocity equal to $(3/4)\text{th}$ of the escape velocity from the surface of the earth. The height it reaches is:
   (A) $\dfrac{10R}{9}$
   (B) $\dfrac{9}{7}R$
   (C) $\dfrac{9}{8}R$
   (D) $\dfrac{10R}{3}$

5. A 0.10 kg block oscillates back and forth along a horizontal surface. Its displacement from the origin is given by: $x=(10\text{cm})\cos\left[(10rad/s)t+\pi/2rad\right]$. What is the maximum acceleration experienced by the block:
   (A) $10\,m/s^2$

(B) $10\pi\ \text{m/s}^2$

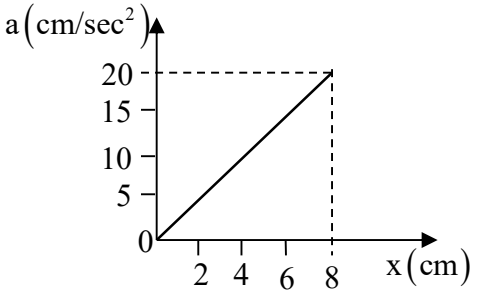
(C) $\frac{10\pi}{2}\ \text{m/s}^2$

(D) $\frac{10\pi}{3}\ \text{m/s}^2$

6. A small disc of radius 2cm is cut from a disc of radius 6cm. If the distance between their centres is 3.2 cm, what is the shift in the centre of mass of the disc?
(A) 0.4 cm
(B) 2.4 cm
(C) 1.8 cm
(D) 1.2 cm

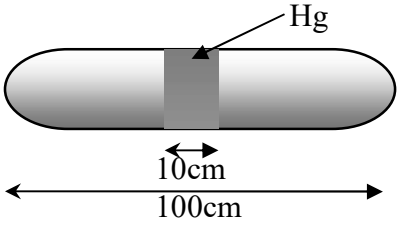
7. Power of a water pump is 4KW. If $g = 10\,\text{m/s}^2$, the amount of water it can raise in one minute to a height of 10 m is:
(A) $2000\,\text{litre}$
(B) $1000\,\text{litre}$
(C) $100\,\text{litre}$
(D) $2400\,\text{litre}$

8. A 5kg mass moves along x-axis. Its acceleration as a function of its position is a shown in the figure. What is the total work done on the mass by the force as the mass moves from $x=0$ to $x=8\,\text{cm}$.



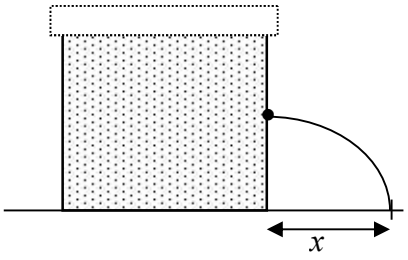
(A) $4\times10^{-2}\text{J}$
(B) $8\times10^{-2}\text{J}$
(C) $2\times10^{-2}\text{J}$
(D) $0.8\times10^{-2}\text{J}$

9. A horizontal uniform glass tube of 100cm, length sealed at both ends contain 10 cm mercury column in the middle. The temperate and pressure of air on either side of mercury column are respectively 80°C and 76cm of mercury. If the air column at one end is kept at 0°C and the other end at 273°C, the pressure of air which is at 0°C is (in cm of Hg):



(A) 76
(B) 68.2
(C) 102.4
(D) 122

10. Water is filled in a fixed container filled upto a height of 13m from ground. A sniper fires and makes a small hole in the side at a height of 9m from the ground. The water jet strikes the ground at X meter from the container. Find the value of X.



(A) 6m
(B) 9 m
(C) 12 m
(D) 13 m

11. A uniform rope of mass 0.1 kg and length 2.45 m hangs from a ceiling. Calculate time taken by a transverse wave to travel the full length of rope:
(A) $0.5\,\text{sec}$
(B) $1\,\text{sec}$
(C) $2\,\text{sec}$
(D) $1.5\,\text{sec}$

12. A siren (point source) creates a sound level of 60 dB at location 500m from the speaker. The siren is powered by a battery that deliver a total energy of 1 kJ. Assuming that efficiency of siren is 30% then the total time for which siren will produce sound is:
(A) $180s$
(B) $95.5s$
(C) $1405s$
(D) $60s$

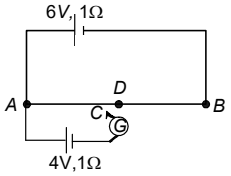
13. In gravity free space Iron Man of mass 'M' standing at height 'h' above the floor throws a Precious Stone of mass 'm' straight down with speed u. When stone reaches the floor, what is the distance of Iron Man above the floor?
(A) $\text{h}\left(1+\frac{\text{m}}{\text{M}}\right)$
(B) $\text{h}\left(1-\frac{\text{m}}{\text{M}}\right)$
(C) $\text{h}\left(\frac{\text{m}}{\text{M}}\right)$
(D) h

14. A black body, initially at temperature T, cools to temperature $\left(\text{T}/2\right)$ in time Δt in surrounding which is near absolute zero. It will cool further to a temperature $\left(\text{T}/4\right)$ in additional time:
(A) $8\Delta\text{t}$
(B) $7\Delta\text{t}$
(C) $9\Delta\text{t}$
(D) None of these

15. A plane mirror reflecting a ray of incident light is rotated through an angle θ about an axis through the point of incidence in the plane of the mirror perpendicular to the plane of incidence, then:
(A) The reflected ray does not rotate
(B) The reflected ray rotates through an angle θ
(C) The reflected ray rotates through an angle 2θ
(D) None of these

16. A force acting on a particle moving in the xy-plane is given by $\text{F}=\left(2y\hat{i}+x^2\hat{j}\right)N$, where x and y are in m. The particle moves along a straight line from the origin to $\left(5,5\right)$. The work done by F is:
(A) 125J
(B) 66.7J
(C) 35J
(D) 25J

17. A 6 volt battery of internal resistance 1Ω is connected across a uniform wire AB of length 100 cm. The positive terminal of another battery of emf 4V and internal resistance 1Ω is joined to the point A as shown. Find the distance of point D from A if there is no current through galvanometer. (resistance of *AB* = 5Ω)



(A) 80 cm
(B) 60 cm
(C) 75 cm
(D) 40 cm

18. Two radioactive materials X_1 and X_2 contain same number of nuclei. If $6\lambda s^{-1}$ and $4\lambda s^{-1}$ are the decay constant of X_1 and X_2 respectively. The ratio of number of nuclei undecayed of X_1 to that of X_2 will be (1/e) after a time:

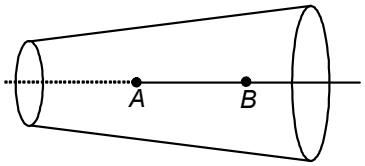
(A) $\frac{1}{2\lambda}\text{sec}$

(B) $\frac{1}{10\lambda}\text{sec}$

(C) $\frac{1}{5\lambda}\text{sec}$

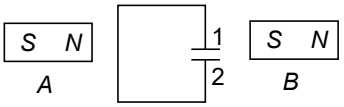
(D) $\frac{1}{\lambda}\text{sec}$

19. A resistance of frustum shape is shown in figure. If a current *i* passes through the resistance, the electric field at *A* and *B* are related as



(A) $E_A > E_B$
(B) $E_B > E_A$
(C) $E_A = E_B$
(D) there is no relation

20. Two bar magnets moving with same speed in the given figure. Choose correct statement.

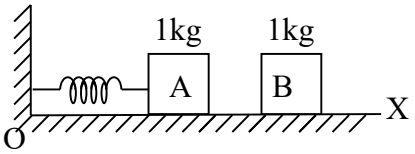


(A) Plate (1) will be +ve relative to (2) if *A* is *B* is moving towards left.
(B) Plate (1) will be +ve relative to (2) if *A* is moving towards left and *B* is moving towards right.
(C) Charge on capacitor increase if *A* and *B* are at rest.
(D) None of these

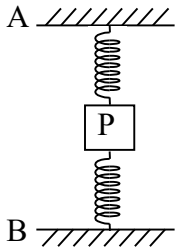
SECTION – B
(Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

21. One end of an ideal spring is fixed to a wall at origin O and axis of the spring is parallel to x-axis. A block of mass 1kg is attached to the free end of the spring and is performing SHM. Equation of position of the block in co-ordinate system shown in figure is x = 10 + 3 sin (10t), where t is in second and x in cm. Another identical block B, moving towards the origin with velocity 0.6 m/s collides elastically with the block A at t = 0. The new amplitude of oscillations will be: (in meter)



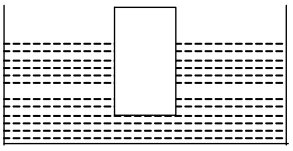
22. Two identical springs are attached to a small block P. The outer ends of the springs are fixed at A and B such that AB is vertical. When P is in equilibrium the extension of top spring is 20 cm and extension of bottom spring is 10 cm. If the period of small vertical oscillations of P about its equilibrium position is $\frac{\pi}{N\sqrt{2}}\sec$, then fill the value of N.



23. A particle is projected at an angle to the horizontal from a point on a smooth horizontal floor. It is found that ratio of the first maximum height attained to maximum height reached by it after 3rd collision with the floor is $\frac{10^{12}}{2^{36}}$. Find coefficient of restitution between particle and floor.

24. A physical quantity $'x'$ is calculated from the relation $x=\frac{a^2b^3}{c\sqrt{d}}$. If percentage error in a, b, c and d are 2%, 1%, 3% and 4% respectively, what is percentage error in $'x'$.

25. A cylinder of mass m and cross section area a is floating partially immersed in a liquid in a beaker of cross section of area A as shown in the figure. Density of the liquid is ρ. Find angular frequency ω of small amplitude of vertical oscillations of the cylinder.

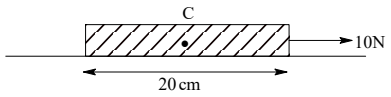


(Take $A=1m^2, a=\frac{1}{2}m^2, \rho=1000kg/m^3, m=\frac{1}{2}kg, g=10m/s^2$ and $\sqrt{2}=1.414$)

26. If a charge q is uniformly spread on a thin dielectric square plate the electric potential at its centre is found to be V_1. If six such charged plates are joined to make a hollow cube, the potential at the center of the cube is found to be V_2. The potential at one of the vertex of this cube is aV_1+bV_2, then find the value of $a+b$? In all the cases, potential at infinity is assumed to be zero.

27. Two satellites of the earth move in a common plane along circular orbits, the radii being r and $r-\Delta r\left(\Delta r << r\right)$. What is the time interval between their periodic approaches to each other over the minimum distance? Take M_c to the mass of the earth.
$\left(M_c = 6\times10^{24}\,kg, r = 7000km, \Delta r = 70km\right)$

28. A force of 10 Newton is applied on a rope of mass 2kg and length $20\,\text{cm}$ which is on a rough table as shown in the figure. Find the tension in the rope at its mid point $\left(\mu=3/4\right)$



29. A closed circular wire hung on a nail in a wall undergoes small oscillations of amplitude 2 and time period 2s. Find the radius (in meter) of the wire?

30. A proton is bombarded on a stationary lithium nucleus. As a result of the collision two $\alpha-$particles are produced. If the direction of motion of the $\alpha-$particles with the initial direction of motion makes an angle $\cos^{-1}\left(1/4\right)$, find the kinetic energy of the striking proton. Given binding energies per nucleon of Li^7 and He^4 are 5.60 and 7.06 MeV respectively? (Assume mass of proton $\approx$ mass of neutron).

Chemistry

PART – B

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

31. Which of the following statement is **INCORRECT** about the homogeneous reaction? $aA + bB \rightarrow cC + dD$.

(A) The relation $-\frac{1}{a}\frac{dn_A}{dt} = \frac{1}{c}\frac{dn_C}{dt}$ need not hold if the reaction has more than one step and concentration of reaction intermediates are not small.

(B) The rate of reaction $= k[A]^a[B]^b$ if the above reaction is elementary reaction.

(C) The rate of reaction = $-\frac{1}{a}\frac{d[A]}{dt} = -\frac{1}{b}\frac{d[B]}{dt} = +\frac{1}{c}\frac{d[C]}{dt} = +\frac{1}{d}\frac{d[D]}{dt}$.

(D) If concentration of A reduces to half in time x seconds than concentration of B reduces to half in time $\frac{b}{a}x$ seconds regardless of initial concentrations of A & B and order of reaction w.r.t. A & B

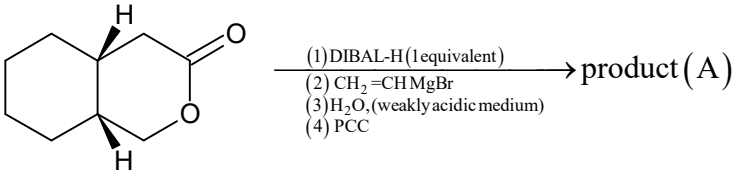
32. Which species is linear at the central atom?

(A) CO_3^{2-}

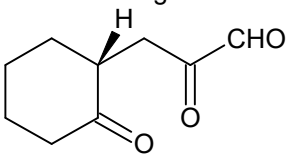
(B) $[S_3\ Sn\ S\ Sn\ S_3]^{6-}$

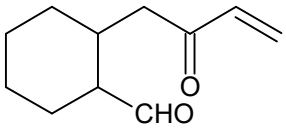
(C) $Me_3SiOSiMe_3$

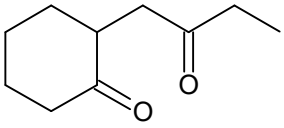
(D) SC_3S

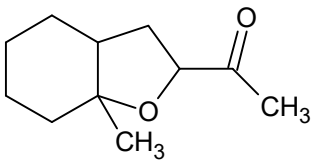
33. 

Which of the following is correct structure of 'A' ?

(A) 

(B) 

(C) 

(D) 

34. Which of the following statement is correct about various periodic properties?
(A) The electron affinity of an atom under consideration is equal to the negative of the electron gain enthalpy at any temperature.
(B) The electronegativity of any given element is not constant; it varies depending on the element to which it is bound.
(C) The first ionization enthalpy values of group 13 elements varies as $\mathrm{B} > Al > \mathrm{Ga} > \mathrm{In} > Tl$.
(D) An element with ground state electronic configuration $[Xe]4f^{14}5d^{1}6s^{2}$ belongs to d-block in the modern periodic table.

35. Which of the following statement is correct about various properties?
(A) At 1 atm pressure boiling temperature is called standard boiling point while at 1 bar pressure boiling point is called normal boiling point.
(B) Standard boiling point of the liquid is slightly greater than the normal boiling point.
(C) Boiling does not occur when liquid is heated in a closed vessel.
(D) Vapour pressure of various compounds at 300K is in order, diethyl ether $> \mathrm{CCl_4} > \mathrm{H_2O} >$ ethyl alchol

36. Calculate bond enthalpy (in kJ/mol) of C-Cl in $\mathrm{CCl_4}(g)$ from the following Data:

$\Delta_{vap.}H^{\circ}_{CCl_4} = 30.5\,kJ/mol \quad \Delta_f H^{\circ}_{(CCl_4,l)} = -135.5\,kJ/mole$

$\Delta_a H^{\circ}(C) = 715\,kJ/mol \ (i.e.\ enthalpy\ of\ atomisation)$

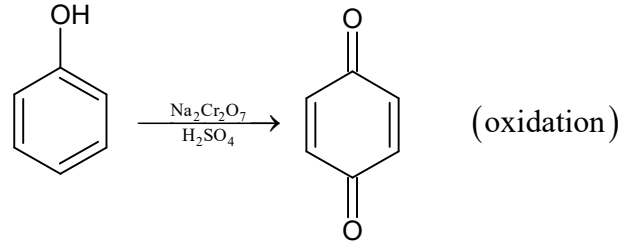
$\Delta_a H^{\circ}_{(Cl_2)} = 242\,kJ/mole$

(A) 326
(B) 333.625
(C) 356.5
(D) 447

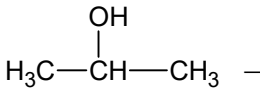
37. For a system (such as an atom or a molecule whose energy does not change with time) the Schrodinger equation is written as $\hat{H}\psi = E\psi$. Which of the following is **INCORRECT** about the Schrodinger equation?
(A) $\hat{H}$ is a mathematical operator called Hamiltonian.
(B) The solution of this equation gives E and ψ.
(C) Schrodinger gave a recipe of constructing $\hat{H}$ operator from the expression for the total energy of the system.
(D) The total energy of the system takes into account the kinetic energies of all the sub-atomic particles, attractive potential between the electrons and nuclei but excludes repulsive potential among the electrons and nuclei individually.

38. Aqueous solution of boric acid is treated with salicylic acid. Which of the following statement is correct for the product formed in the above reaction?
(A) No product will be formed because both the acid
(B) Product is 4-coordinated complex and optically resolvable
(C) Product is 6-coordinated complex and optically non-resolvable
(D) There are two rings only which are five membered

39. Which of the following reaction **INCORRECTLY** matched?

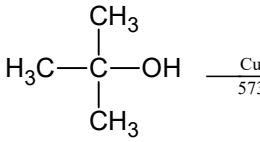
(A) 

(oxidation)

(B) 

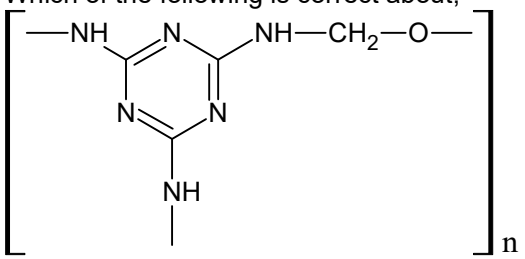
(oxidation)

(C) $CH_3-CH=CH-CH_2-OH \xrightarrow{PCC} CH_3-CH=CH-CHO$ (dehydrogenation)

(D) 

(oxidation)

40. Which of the following is correct about,



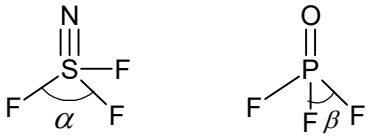
polymer?

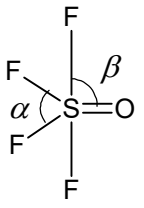
(A) Polymer is formed by addition polymerisation of melamine and formaldehyde.
(B) It is used in manufacturing of unbreakable crockery.
(C) It is a thermoplastic polymer i.e. capable of repeatedly softening on heating and hardening on cooling.
(D) It is a linear polymer.

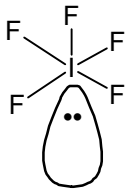
41. Which of the following statement is correct about antimicrobials?
(A) Ampicillin and amoxycillin are narrow spectrum antibiotics.
(B) Antibiotics have both cidal (killing) effect and static (inhibitory) effect on microbes.
(C) A substance cannot act as an antiseptic as well as disinfectant by varying the concentration.
(D) A mixture of chloroxylenol and terpineol, tincture of iodine, iodoform, boric acid etc are used as antiseptics.

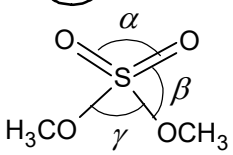
42. Which of the following Vitamin is not matched with its correct scientific name?
(A) Vitamin B_1=Threonin
(B) Vitamin B_2=Riboflavin
(C) Vitamin B_6=Pyridoxine
(D) Vitamin C=Ascorbic acid

43. Which of the following properties about bond angle and bond lengths is correct for the given compound?

(A)  $\alpha > \beta$

(B)  $\alpha = 120°$ *and* $\beta > 90°$

(C)  all I-F bonds are of equal length because Iodine is sp^3d hybridized.

(D)  $\alpha > \beta > \gamma$

44. Calculate the molar solubility of $Ni(OH)_2$ in 0.1 M NaOH . The solubility product of $Ni(OH)_2$ is 2×10^{-15} . (Given: $\sqrt[3]{0.5} = 0.79$)

(A) 2×10^{-13}M

(B) 2×10^{-14}M

(C) 7.9×10^{-6}M

(D) 7.9×10^{-4}M

45. Calculate ΔG for the following reaction

$$H_2O(l) \longrightarrow H_2O(g)$$
$$(2atm, 373K) \quad (2\,atm, 373K)$$

(Given: $0.0821\times373\times\ln 2 = 21.23$ atm-ltr, molar volume of water at 2 atm , 373 K=18ml)

(A) $\Delta G = 0$

(B) $\Delta G > 0$

(C) $\Delta G < 0$

(D) None of these

46. Which of the following is not a magnesium ore?

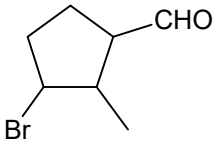
(A) Carnalite

(B) Dolomite

(C) Magnesite

(D) Malachite

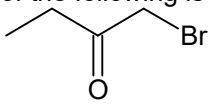
47. Which of the following is the correct IUPAC name of the compound?

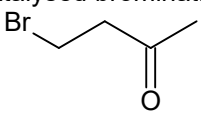


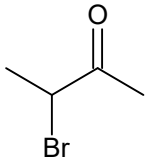
(A) 2-methyl-3-bromocyclopentane carbaldehyde
(B) 1-formyl-2-methyl-3-bromocyclopentane
(C) 3-bromo-2-methyl cyclopentane carbaldehyde
(D) 1-bromo-2-methyl cyclopentane-3-carbaldehyde

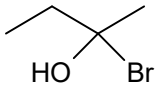
48. Which of the following is correct about defects in solid crystals?
(A) Vacancy defects and interstitial defects are two main types of non-stoichiometric defects.
(B) If molten NaCl containing a little amount of MCl_3(ionic) is crystallised, the cationic vacancies thus produced are equal to twice the number of M^{3+} ions.
(C) Zinc oxide is white in colour at room temperature, on heating it turns green and its formula becomes $Zn_{1-x}O$
(D) Frenkel defect is shown by ionic substance in which the cation and anion are of almost similar sizes.

49. Which of the following is the major product in the acid catalysed bromination of butan-2-one?

(A) 

(B) 

(C) 

(D) 

50. A sulphide of a metal (M) with formula MS is white coloured ppt which is soluble in acidic solution $(PH<6)$. Metal cation M is:
(A) Pb^{2+}
(B) Zn^{2+}
(C) Hg^{2+}
(D) Cu^{2+}

SECTION – B
(Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

51. In a FCC structure of edge length 8Å, the area of triangle formed by line joining two nearest tetrahedral void to their nearest octahedral void is X$Å^2$ then value of [X] is ___.

52. Among the following, number of compounds that would require electrolysis process to get their respective metal is……..
$NaCl, Cr_2O_3, MgCl_2, Al_2O_3, CaCl_2, Fe_2O_3$

53. When $6\times10^{-5}\text{g}$ of a protective colloid was added to 20 mL of a standard gold sol, the precipitation of latter was just prevented on addition of 1 mL of 10% NaCl solution. The gold number of the protective colloid is found to be $x\times10^{-2}$. The value of x is.

54. Given that crystal field stabilization energy for $\left[\text{Co}\left(\text{H}_2\text{O}\right)_6\right]^{2+}$ is 7200 cm^{-1}, pairing energy for Co^{2+} is $20,800\,cm^{-1}$, then the value of $\frac{\Delta_0}{10^3}\left(\text{in cm}^{-1}\right)\text{is}:$

55. NH_3 reacts with a gas produced when methanol is passed through Cu tube at 573K. The compound obtained is polycyclic. How many rings (6 members) does it contain = [X]. The number of Nitrogen atoms it has = [Y]. The value of x+y is:

56. Two moles of an ideal gas (C_V = R) was compressed adiabatically against constant pressure of 2 atm. Which was initially at 350 K and 1 atm pressure and final pressure=8/3 atm. If the work involve in the process is equal to 100n × R. Then what will be the value of n.

57. A buffer solution is formed by mixing 100 mL 0.01 M CH_3COOH with 200 mL 0.02 M CH_3COONa. If this buffer solution is made to 1 lit by adding 700 mL of water, ratio of final pH to initial pH:

58. In rusting of iron, iron is oxidized and O_2 is reduced. The no. of electrons used during reduction of 1 mole of O_2 are …

59. The vapour pressure of a mixture of two volatile liquids 'A' & 'B'; is expressed as $P_{mix}=6.0+1.5X_B$, where X_B is the mole fraction of B in the mixture. The vapour pressure of 'A' at the same temperature is _____.

60. A mixture of N_2 and H_2 in the molar ratio 1 : 3 attains equilibrium when 50% of mixture has reacted. If P is the total pressure of the mixture, the partial pressure of NH_3 formed is P/y. The value of y is:

Mathematics

PART – C

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

61. Which of the following function is surjective but not injective?

(A) $f : R \to R,\ f(x) = x^4 + 2x^3 - x^2 + 1$

(B) $f : R \to R,\ f(x) = x^3 + x + 1$

(C) $f : R \to R^+,\ f(x) = \sqrt{1 + x^2}$

(D) $f : R \to R,\ f(x) = x^3 + 2x^2 - x + 1$

62. If $[x]$ denotes the integral part of x and in $(0, \pi)$, we define

$$f(x) = \begin{cases} \left[\dfrac{2(\sin x - \sin^n x) + |\sin x - \sin^n x|}{2(\sin x - \sin^n x) - |\sin x - \sin^n x|}\right], & x \neq \dfrac{\pi}{2} \\ 3, & x = \dfrac{\pi}{2} \end{cases}$$

Then for $n > 1$,

(A) $f(x)$ is continuous but not differentiable at $x = \dfrac{\pi}{2}$

(B) both continuous and differentiable at $x = \dfrac{\pi}{2}$

(C) neither continuous nor differentiable at $x = \dfrac{\pi}{2}$

(D) $\lim\limits_{x \to \frac{\pi}{2}} f(x)$ exists but $\lim\limits_{x \to \frac{\pi}{2}} f(x) \neq f\left(\dfrac{\pi}{2}\right)$

63. Let f be a function defined on R by $f(x) = [x] + \sqrt{x - [x]}$, then

(A) f is not continuous at every $x \in I$

(B) f is not continuous at every $x \in R \sim I$

(C) f is a continuous function

(D) None of these

64. The maximum value of area of rectangle $OABC$, where O is origin, A lies on positive y-axis and B lies on the curve $y = \dfrac{\ln x}{x^2}$, is

(A) e^{-1}

(B) $e^{-1/2}$

(C) 1

(D) $e^{1/2}$

65. Let $f(x)=\int \frac{x^2 dx}{(1+x^2)(1+\sqrt{1+x^2})}$ and $f(0)=0$. Then *f* (1) is

(A) $\log(1+\sqrt{2})$

(B) $\log(1+\sqrt{2})-\frac{\pi}{4}$

(C) $\log(1+\sqrt{2})+\frac{\pi}{4}$

(D) none of these

66. If statement $(p \to q) \to (q \to r)$ is false, then truth values of statement p, q, r respectively, can be-

(A) FTF
(B) TTT
(C) FFF
(D) FTT

67. Area bounded by $y = \tan^{-1}x$, $y = \cot^{-1}x$ and *y*-axis is equal to

(A) $\ln \sqrt{2}$ sq. units

(B) $\ln$ 4 sq. units

(C) $\ln$ 8 sq. units

(D) $\ln$ 2 sq. units

68. Solution of $\sec^2 y\frac{dy}{dx}+x\tan y=x^3$ is

(A) $\tan y = x^2 + ce^{-x^2}$

(B) $\tan y = x^2 - 2 + ce^{-x^2}$

(C) $\tan y = x^2 - 2 + ce^{-x^2/2}$

(D) none of these

69. If $y=\frac{\cos 6x+6\cos 4x+15\cos 2x+10}{\cos 5x+5\cos 3x+10\cos x}$, then $\frac{dy}{dx}$ is equal to

(A) $2\sin x+\cos x$

(B) $-2\sin x$

(C) $\cos 2x$

(D) $\sin 2x$

70. The tangents at z_1, z_2 on the circle $|z-z_0|=a$, meet at z_3. Then $\left(\frac{z_3-z_1}{z_0-z_1}\right)\left(\frac{z_0-z_2}{z_3-z_2}\right)$ is equal to

(A) 0
(B) 1
(C) –1
(D) 2

71. The number of permutations of letters of word HINDUSTAN such that neither the pattern 'HIN' nor 'DUS' nor 'TAN' appears are
(A) 12246
(B) 169194
(C) 166680
(D) 181434

72. *A* is a 3 × 3 matrix with entries from the set {–1, 0, 1}. Then the probability that *A* is neither symmetric nor skew-symmetric is
(A) $\frac{3^9-3^6-3^3+1}{3^9}$
(B) $\frac{3^9+3^6-3^3+1}{3^9}$
(C) $\frac{3^9-3^6+3^3+1}{3^9}$
(D) None of these

73. Number of value(s) of *x* satisfying the equation $3x^2-2x^3=\log_2\left(x^2+1\right)-\log_2 x$ is
(A) 1
(B) 2
(C) 3
(D) None of these

74. If $(1+x+x^2)^n=\sum_{r=0}^{2n}a_r x^r$, then the value of $\sum_{k=0}^{3r}(-1)^k a_k\ {}^nC_{3r-k}$ (where $3r\le n$) equals
(A) $(-1)^r.{}^nC_r$
(B) ${}^nC_{3r}$
(C) $(-1)^{3r}.{}^nC_{3r}$
(D) ${}^nC_r$

75. If $x\in R$ and $n\in I$, then the determinant $\Delta=\begin{vmatrix}\sin(n\pi) & \sin x-\cos x & \log\tan x\\ \cos x-\sin x & \cos\left[(2n+1)\frac{\pi}{2}\right] & \log\cot x\\ \log\cot x & \log\tan x & \tan(n\pi)\end{vmatrix}$ equals
(A) 0
(B) $\log\tan x-\log\cot x$
(C) $\tan\left(\frac{\pi}{4}-x\right)$
(D) None of these

76. If $A=\begin{bmatrix}2 & 3 & 0\\ 1 & \sin^2 x & -\cos^2 x\\ \cos^2 x & \cos^4 x & \sin^4 x\end{bmatrix}$ and $\mathrm{A\,(Adj\,A)=K}\begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix}$, then the value of K is
(A) 0
(B) –1
(C) 1
(D) None of these

77. If *m* is the root of the equation $(1-ab)x^2-(a^2+b^2)x-(1+ab)=0$ and *m* harmonic means are inserted between *a* and *b*, then the difference between the last and the first of the means equals

(A) $ab(a+b)$

(B) $a(b-a)$

(C) $ab(b-a)$

(D) $b-a$

78. If the tangents at two points (1, 2) and (3, 6) on a parabola intersect at the point (1, 6), then equation of directrix is

(A) $2x+y-8=0$

(B) $x-2y+11=0$

(C) $x+2y-13=0$

(D) $2x-y+4=0$

79. The equation of the circle, which is touched by the line $y = x$, has its centre on the positive direction of the *x*-axis and cut off a chord of length 2 units on the line $\sqrt{3}\ y=x,$ is

(A) $x^2+y^2-4x+2=0$

(B) $x^2+y^2-4x+1=0$

(C) $x^2+y^2-8x+8=0$

(D) $x^2+y^2-4y+2=0$

80. If the equal sides *AB* and *AC* along positive *x* and *y* axes (each equal to *a*) of a right angled isosceles triangle *ABC* be produced to *P* and *Q* respectively so that *B*P. *CQ* = AB^2, then the line *PQ* always passes through a fixed point

(A) (*a*, 0)

(B) (0, *a*)

(C) (*a, a*)

(D) none of these

SECTION – B
(Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

81. If p, q, r, s are the probabilities of raining at four different places at some fixed moment, then the maximum value of $3(p^2 + q^2 + r^2 + s^2) – 2(p + q + r + s) + 4$ is _____

82. Let f(x), $x \geq 0$ be a non-negative continuous functions. If f′(x) $\cos x \leq f(x)\sin x, \forall x \geq 0,$ then $f\left(\frac{5\pi}{3}\right)$ is _____

83. The value of $\lim\limits_{n\to\infty}\sum\limits_{r=0}^{n}\frac{{}^nC_r}{n^r(r+3)}$ = e – x, then x is _____

84. Let f(x) be a polynomial of degree 4 such that f([i]) = $\frac{1}{1+[i]}$ for $0 \le i \le 4$, where [x] = Greatest integer function, then f(6) = _____

85. $A=\begin{bmatrix}1 & 2k-1\\ 0 & 1\end{bmatrix}$, then the value of $\prod_{k=1}^{50}\begin{bmatrix}1 & 2k-1\\ 0 & 1\end{bmatrix}$ is equal to $A=\begin{bmatrix}1 & r\\ 0 & 1\end{bmatrix}$ then r is equal to 500K. Then K is _____

86. If a, b, c are 3 positive real numbers, then minimum value of $\frac{a^2+1}{b+c}+\frac{b^2+1}{c+a}+\frac{c^2+1}{a+b}$ is_____.

87. Let S_n denote the sum to *n* terms of an arithmetic progression. If the common difference is equal to $S_n - kS_{n-1} + S_{n-2}$, then *k* =

88. If tangent and normal at point *P* (in first quadrant) to ellipse $\frac{x^2}{25}+\frac{y^2}{16}=1$ intersect major axis at *T* and *N* respectively in such a way that ratio of area of ΔPTN and $\Delta PSS'$ is $\frac{91}{60}$, $\text{ar}\left(\Delta PSS'\right)\times\sqrt{3}$ is (*S* and *S*′ are foci)

89. Standard Deviation of a data is 7, when each observation is decreased by 7 then standard deviation of new data is____

90. For $x>-1$, let $f(x)=\int_0^{\pi/4}\log_e(1+x\tan z)dz$. If the value of $f\left(\frac{1}{2}\right)-f\left(\frac{1}{3}\right)$ is $\frac{\pi}{m}\log_e\left(\frac{n}{k}\right)$ where n,k are coprimes; m,n,k then value of $m+n+k$ is_____